Summer 2026 Production-Development Plan for the OAA Self-Aware Room (SAR) / LG-038 Installation

Abstract

This document defines the Summer 2026 production-development plan for the OAA Self-Aware Room (SAR) / LG-038 installation, with a specific focus on building the physical and media infrastructure needed before more advanced sensing and AI integration are attempted. The summer work is framed as the development of a modular blended-environment platform: standardized 8020-based “intelligent flats” that can assemble into triptych projection surfaces, support front and rear projection testing, accommodate masking, carry local power distribution, and reserve mounting capacity for speakers, cameras, sensors, and future computational systems. These triptychs will be developed not as isolated scenic objects, but as modular media-capable architectural units within the larger room ecology, which includes immersive audio infrastructure, projector/camera systems, and the longer-term SAR research agenda.

The document subdivides the summer activity into practical task areas: structural design and fabrication of the 8-foot triptychs; fabric attachment and tensioning tests; projection fabric evaluation for front/rear projection, acoustic transparency, and camera capture; masking and visual-continuity strategies; power and cable-routing design; possible localized speaker integration within each flat; camera/projection geometry; room-level documentation; and student researcher roles. A Gantt-style roadmap beginning around July 10 organizes the summer into phases: orientation and existing-condition documentation; 6-foot prototype testing; standardized flat design; 8-foot triptych fabrication; projection/audio/camera integration; and final documentation. The final deliverable is not simply a completed scenic object, but a reusable installation package: CAD drawings, assembly procedures, test matrices, fabric/projection findings, camera and speaker placement notes, power-routing diagrams, and recommendations for fall development.

1. Project Context

The OAA Self-Aware Room (SAR) project is a modular blended-environment research initiative centered on LG-038 and situated within the larger BBS/BRPS meta-project ecosystem. Its purpose is to develop a persistent room-scale platform in which physical structures, projection surfaces, cameras, immersive and localized sound, digital-twin representations, and future sensing/AI systems can be configured, studied, and refined as parts of a coherent environment. SAR should not be understood as a conventional smart classroom with fixed commercial automation. It is better understood as a construction-kit research environment: a deliberately modular system whose components can be assembled into specific configurations, tested, documented, standardized, and then recombined for later research, production, teaching, and performance uses.

The project is now moving from speculation and precedent research into design and production development. Purchase requests have been submitted through purchasing, with at least one purchase request already moving forward. Because the availability and arrival date of equipment will affect the order of installation, the summer plan should remain dependency-aware rather than assuming that all systems will be available at the start of the work period. Some baseline work has already occurred: the room and existing prototype have been measured. The next planning task is therefore not measurement from scratch, but integration of those measurements into the project's GitHub documentation so that they can guide fabrication, digital-twin development, and future student work.

LG-038 is being developed as a persistent blended research environment rather than a conventional classroom, laboratory, or performance venue. Its value comes from the interaction among physical structures, virtual media, computational systems, sound, projected imagery, camera capture, and human participants. The immediate summer priority is not to complete a fully self-aware room in the computational sense, but to establish the room's physical and mediated substrate: the surfaces, frames, projection systems, audio infrastructure, power pathways, camera/projector relationships, and documentation practices that make later intelligent behavior possible.

The triptych screen system is one of the first major physical components in this environment. Because the triptychs will define visual surfaces, spatial boundaries, possible audio-source locations, and future sensor-mounting zones, their design should be treated as foundational infrastructure. The project builds on an existing 6-foot prototype triptych and will move toward taller 8-foot triptychs constructed from 8020 / T-slot materials. The initial prototype should remain an active test article, but the summer schedule should move quickly toward construction of at least one 8-foot flat during the first couple of weeks. This early 8-foot flat will serve as a revised working prototype for puppeteers and production researchers, allowing the team to test scale, handling, projection behavior, masking, and puppet interaction under conditions closer to the intended final system.

The summer work also includes several parallel project activities. The team is testing layered video imagery as a way to simulate shadow-casting onto a central screen. This can proceed in parallel with fabric attachment, projection-surface testing, and triptych masking studies, since the results will affect how the central projection surface is used. The team is also investigating puppet fabrication using purchased materials that are awaiting fabrication and construction. These puppet-material studies should be documented as part of the same production-development ecosystem, since puppet surfaces, projection behavior, shadow behavior, and camera capture all intersect within the larger blended-performance research agenda.

A further parallel activity involves reviewing recordings of past performances to determine whether they can be repurposed or transformed into prerecorded projection material for the main screen. This should be treated as both a media-production task and a mediation-pathway experiment: a live performance is captured, edited, transformed, and reprojected into the room as a new spatial event. This activity can generate useful test media for the screen system even before new live performances occur.

The digital twin of LG-038 is another major component of the larger project. Although the summer may not complete a fully interactive room model, the physical installation work should be documented in ways that support the digital twin: accurate room measurements, triptych dimensions, equipment locations, speaker positions, projector/camera positions, cable pathways, and evolving room states. The digital twin should be understood as a coordinating representation of the room rather than a separate side project.

The immersive audio system installation and configuration is also part of the summer production-development environment. The room should be described as an immersive audio research environment that includes a Dolby Atmos-compatible 7.1.4 configuration, while also supporting distributed-speaker and localized-source approaches. This means the triptychs, puppet systems, projection surfaces, and digital twin should all be considered in relation to the room's sound field and future spatial-audio experiments.

2. Ultimate Goal and Summer Subprojects

The ultimate goal is to develop LG-038 into a persistent self-aware blended-environment research room: a configurable physical, virtual, conceptual, visual, auditory, and computational space in which media surfaces, immersive audio, cameras, projection systems, digital-twin representations, and future sensing/AI systems can operate as an integrated research platform. The summer work will not necessarily complete this full goal. Instead, it should establish the modular physical and media infrastructure that allows the room to move toward that goal in a disciplined and reusable way.

The summer activity can be organized as a sequence of overlapping subprojects. These are presented in roughly chronological order, though several will run in parallel depending on equipment arrival, room access, and personnel availability.

- **Equipment acquisition and procurement tracking.**
  The project is currently in the equipment-acquisition phase, with purchase requests submitted through purchasing. The hoped-for outcome is a clear procurement status log that identifies what has arrived, what is pending, what has dependencies, and what can be installed or tested during the summer.

- **GitHub integration of existing measurements.**
  The prototype triptych and LG-038 room have already been measured. The hoped-for outcome is a GitHub-based measurement package that includes room dimensions, prototype dimensions, photographs, diagrams, naming conventions, and enough contextual explanation for future students and collaborators to use the information reliably.

- **Existing 6-foot prototype documentation and continued testing.**
  The existing prototype should remain available as a test rig for fabric attachment, projection behavior, masking studies, and camera/projection experiments. The hoped-for outcome is a documented record of what the prototype already demonstrates and what must change in the 8-foot version.

- **Construction of the first 8-foot flat.**
  One revised 8-foot flat should be built early in the summer so puppeteers and production researchers can begin working with the new scale. The hoped-for outcome is a usable working prototype that tests handling, stability, projection scale, masking assumptions, and puppet interaction before the full triptych system is completed.

- **Standardized intelligent flat design.**
  The first 8-foot flat should lead to a repeatable flat standard: a modular 8020-based frame that can support projection fabric, masking, power routing, possible localized speakers, future sensors, and hinge/pivot connections. The hoped-for outcome is a documented flat standard that can be duplicated and combined into triptychs.

- **Fabric attachment and tensioning research.**
  The project must determine how projection fabric attaches to the frame and how it can be tensioned easily, correctly, and repeatably. The hoped-for outcome is a preferred attachment strategy, such as a near-front inset plane, clamp-bar system, rod/sleeve system, or other repeatable method, with documentation sufficient for later reconstruction.

- **Projection fabric testing.**
  Candidate fabrics should be evaluated for front projection, rear projection, brightness, diffusion, hotspotting, wrinkle behavior, shadow behavior, camera capture, and acoustic transparency. The hoped-for outcome is a fabric test matrix that identifies which materials are most appropriate for triptych use and which require further testing.

- **Masking and visual-continuity design.**
  The triptych screens should appear as connected as possible, even though structural framing is necessary. The hoped-for outcome is a standardized masking strategy that hides or softens frame edges, hinges, cable paths, and fabric attachment hardware while preserving modularity.

- **Layered video and simulated shadow-casting tests.**
  The project will test whether layered video imagery can simulate shadow-casting onto a central screen. The hoped-for outcome is a set of projection tests and media examples that show whether prerecorded, generated, or composited imagery can support the visual language of the Blended Shadow Puppet work.

- **Prerecorded performance projection tests.**
  Recordings of past performances should be reviewed to determine whether they can be transformed into projected material for the main screen. The hoped-for outcome is a set of candidate excerpts and projection tests that convert past performance documentation into new spatial media for the room.

- **Puppet fabrication, material studies, and puppet-stage module.**
  Purchased materials awaiting fabrication should be used to test puppet construction methods, translucency, opacity, edge quality, durability, joint behavior, and projection/shadow response. In parallel, the project should develop the concept of a self-contained puppet-stage module that attaches to one of the standardized panels, creating the depth needed for puppets to operate either in front of or behind the projection screen. This module should also consider an organza roll or similar soft-goods system that can introduce pigmented materials, building on the second-year production approach. The hoped-for outcome is a documented puppet-material and puppet-stage module study that can guide future BSP fabrication and performance testing.

- **Immersive audio installation and configuration.**
  The room's sound system should be installed and configured as an immersive audio research environment that includes a Dolby Atmos-compatible 7.1.4 configuration while also supporting distributed-speaker and localized-source approaches. The hoped-for outcome is an operational audio infrastructure with documentation of speaker positions, configuration assumptions, and future integration paths.

- **Localized speaker planning for triptychs.**
  Each flat or triptych may eventually support a small local speaker, possibly top-mounted or behind an acoustically transparent fabric. The hoped-for outcome is not necessarily final installation, but a design provision for speaker mounting, cable routing, and acoustic-transparency evaluation.

- **Camera and projection geometry.**
  The relationships among projectors, screens, cameras, puppeteers, and viewers should be tested and documented. The hoped-for outcome is a repeatable set of placement notes, diagrams, and test media that reduce reliance on improvised setup.

- **Digital twin development for LG-038.**
  The physical installation should be documented in ways that support the evolving digital twin of the room. The hoped-for outcome is a digital-twin update package containing dimensions, equipment positions, triptych locations, speaker locations, camera/projector positions, and possible room states.

- **End-of-summer installation package.**
  The summer should conclude with a reusable documentation package rather than

only a collection of built objects. The hoped-for outcome is a GitHub-organized installation package containing CAD drawings, assembly notes, fabric test results, projection tests, audio notes, puppet-material findings, power/cable-routing concepts, digital-twin updates, photographs, process videos, and recommendations for fall continuation.

3. Design Framing: From Scenic Flats to Intelligent Flats

On one level, the triptychs can be understood as scenic flats. They are framed surfaces used to shape visual space. However, within the context of the Self-Aware Room and the broader BBS / BRPS research trajectory, they should be understood as intelligent flats: standardized modular frames that provide a projection surface while also supporting future power, sensing, sound, camera, and computational infrastructure.

Each flat should be treated as a reusable unit. A triptych is then created by connecting three such flats through a hinged or pivoting system. This strategy preserves scenic familiarity while allowing modular expansion. The design goal is to create a flat that can function independently, as part of a triptych, or eventually as part of a larger configurable environment.

The standardized flat should include:

- an 8-foot structural frame based on 8020 / T-slot extrusion;
- a fabric tensioning system suitable for projection textiles;
- a masking attachment strategy;
- provision for front and/or rear projection;
- a top or rear speaker-mounting option;
- low-voltage and/or power-routing capacity;
- reserved mounting zones for future sensors and small electronics;
- hinge or pivot interfaces for triptych assembly;
- cable-routing and labeling standards;
- documentation sufficient for later reconstruction or modification.

The triptych should therefore be considered a modular media object rather than a fixed scenic backdrop.

4. Major Summer Work Areas

The summer work should be understood as a set of parallel production-development tracks rather than a single linear build. Some activities depend on equipment arrival and room access, while others can proceed immediately through testing, documentation, media preparation, and design refinement. The following work areas should be coordinated so that each produces usable documentation and informs the others.

4.1 Structural Design and Fabrication

The first major work area is the translation from the existing 6-foot prototype to one or more 8-foot triptychs. Because the prototype and room have already been measured, early work should focus on converting those measurements into usable GitHub documentation, CAD diagrams, fabrication notes, and cut-list assumptions. The structural design should preserve modularity and allow future expansion.

The schedule should prioritize construction of one 8-foot flat as an early revised prototype. This will allow puppeteers and production researchers to work with the newer scale before the full triptych system is completed. The first flat should therefore be built not only as a component of the final triptych, but also as a practical rehearsal and testing object.

Important structural questions include frame dimensions, hinge geometry, stability, base design, caster or wheel strategy, pivot behavior, transportation, storage, and the degree to which each flat can remain symmetrical and reversible. The more symmetrical the flat, the more easily it can be used as a left, center, or right panel.

4.2 Fabric Attachment and Tensioning

The fabric attachment method is one of the core design problems for the summer. Projection fabric must remain flat, replaceable, testable, and tensioned evenly. The team should compare several attachment strategies before finalizing the 8-foot design.

The main design alternatives include wrapping the fabric around the frame, rear-mounting the fabric, creating an inset near-front projection plane, or developing a partial inner-frame / clamp-bar system. The current working hypothesis is that the projection fabric should not permanently wrap around the entire 8020 structural frame. Instead, the frame should remain accessible for hinges, power, speakers, masking, and future sensor attachment. A near-front inset fabric plane using clamp bars, rods, sleeves, or another adjustable tensioning method may offer the best balance between visual continuity and modular serviceability.

The 6-foot prototype should be used to test fabric tensioning before committing to the 8-foot version.

4.3 Projection Fabric Research

The project should test multiple fabrics for their suitability as projection surfaces. This research should build on existing analysis from the SeaChange 360 project while adapting those findings to the current triptych architecture.

Each candidate fabric should be evaluated for front projection, rear projection, brightness, diffusion, hotspotting, wrinkle visibility, shadow behavior, camera capture quality, and acoustic transparency. Acoustic transparency is relevant because speakers may be mounted behind the projection surface if the fabric permits acceptable sound transmission.

The result should be a fabric test matrix that documents the strengths and weaknesses of each material in relation to the room's intended uses.

4.4 Masking and Visual Continuity

The triptychs should feel as connected as possible, even though the framing is structurally necessary. Masking should therefore be treated as a formal subsystem rather than an afterthought.

The masking layer may need to hide or soften frame edges, hinge gaps, cable paths, speaker mounts, and differences between projection surfaces. It may also define the perceived projection aperture. The project should investigate removable masking strategies, including black theatrical borders, soft goods, Velcro-mounted strips, magnetic attachments, clamp-on borders, or other reusable methods.

Because the triptychs are modular, the masking should also be modular. The design should allow students and researchers to reconfigure the flats without rebuilding custom masking for every arrangement.

4.5 Audio Integration

The LG-038 sound system should be understood as an immersive audio research environment that includes a Dolby Atmos-compatible 7.1.4 configuration but is not limited to that format. The room must also support distributed-speaker and localized-source approaches, including sound emerging from specific speakers or media objects rather than only from an Atmos convolution field.

For the triptychs, the summer design should consider whether each flat can support a small speaker, likely near the top or potentially behind the projection fabric if the fabric is acoustically transparent. This does not require full speaker integration during the first build, but the frame should reserve physical and electrical capacity for this possibility.

Potential audio-related design tasks include speaker mounting points, acoustic-transparency testing, cable routing, power access, and documentation of how localized triptych speakers might interact with the larger immersive audio system.

4.6 Power and Cable Routing

Each triptych should be considered a potential local infrastructure node. Even if advanced sensors are not integrated this summer, the frame should include pathways for power and signal. Purchased bussing and power-distribution materials suggest that power architecture is already part of the system concept.

The summer work should establish a safe and documented approach to local power distribution, low-voltage routing, cable strain relief, labeling, and access. Any power system should be designed conservatively, with appropriate supervision and review by qualified personnel. student researchers may assist with documentation, layout, labeling, and physical organization, but should not be treated as the lead electrical designer.

4.7 Camera and Projection Geometry

The room requires careful consideration of projector placement, surface geometry, camera placement, and sightlines. The triptychs should be designed so that the relationship among projector, fabric surface, camera, viewer, and performer can be reproduced and documented.

Camera studies may include fixed cameras, webcams, documentation cameras, and possible future sensing cameras. Projection studies should evaluate front and rear projection, projector throw, alignment, edge behavior, and whether the triptych geometry can be reliably repeated.

The goal is to move from improvised setup to documented media geometry.

4.8 Layered Video and Simulated Shadow-Casting

A parallel research-production activity will investigate layered video imagery as a way to simulate shadow-casting onto a central screen. This work can proceed alongside fabric and projection-surface testing because it depends on the same questions of brightness, contrast, diffusion, masking, and camera capture.

The layered-video work should test whether prerecorded or generated imagery can create convincing shadow-like relationships on the screen without requiring every shadow to be produced physically in real time. This is especially relevant for the Blended Shadow Puppet research trajectory, where physical puppets, projected imagery, prerecorded media, live camera feeds, and AI-assisted image generation may all contribute to the final visual field.

Possible tasks include creating test clips, layering silhouettes over projected environments, experimenting with opacity and blur, testing front and rear projection behavior, and documenting which approaches remain visually legible on the selected fabrics.

4.9 Puppet Fabrication, Material Studies, and Puppet-Stage Module

The summer activity also includes investigation of puppet fabrication using purchased materials that are awaiting fabrication and construction. These tests should be treated as part of the same blended-environment system rather than as a separate craft activity.

Puppet-material studies should evaluate translucency, opacity, edge sharpness, durability, attachment methods, scale, joint behavior, projection response, shadow quality, and camera capture. If puppets are intended to operate in front of, behind, or within the triptych environment, their material behavior must be evaluated in relation to the screen fabrics and projection geometry.

In addition to puppet fabrication, the project should explore a self-contained puppet-stage module that can attach to a standardized triptych panel or flat. This module would create the depth needed for puppets to operate either in front of the screen or behind it, depending on the projection and shadow-casting strategy. The module should be conceived as an add-on component rather than a permanent redesign of the flat, preserving the modularity of the triptych system.

The puppet-stage module should also consider an organza roll or similar soft-goods mechanism that can introduce pigmented material into the visual field. This builds on the second-year production approach, where pigmented organza or related materials contributed to the shadow/projection language. A roll-based system could allow the team to vary opacity, color, texture, and diffusion without replacing the main projection fabric.

The outcome should be a documented set of puppet fabrication findings and a preliminary puppet-stage module concept, including attachment logic, depth requirements, soft-goods options, and interaction with projection, masking, and camera capture.

4.10 Prerecorded Performance Projection

Another parallel activity will examine whether recordings of past performances can be transformed into projection material for the main screen. This may include editing, masking, compositing, scaling, color correction, shadow simulation, and projection testing.

This work has immediate practical value because it can provide test content for the screen system before new live performances occur. It also has conceptual value within BBS because it transforms a past live event into recorded media, then into a projected spatial event within the room. That transformation should be documented as a mediation pathway involving capture, storage, editing, playback, projection, and audience perception.

4.11 Digital Twin of LG-038

The digital twin of LG-038 should be developed in parallel with the physical installation. The digital twin does not need to be complete before installation begins; rather, it should evolve as the room evolves.

At minimum, the summer work should support the digital twin through accurate measurements, equipment placement records, triptych dimensions, speaker locations, projector/camera locations, cable-routing notes, and photographic documentation. If possible, the digital twin should include multiple room states, showing how triptychs and other modular elements can be reconfigured.

The digital twin should eventually function as a planning, documentation, simulation, and communication tool for the Self-Aware Room (SAR) project.

4.12 Documentation, GitHub Integration, and Knowledge Capture

The installation process itself is research data. The summer project should produce structured documentation that allows future students, faculty, and collaborators to understand what was built, why design decisions were made, and how the system can be extended. Since the prototype and room have already been measured, those existing measurements should be moved into the GitHub site early in the summer. This should include clear file organization, naming conventions, version notes, and enough explanatory context for later students to use the measurements without relying on oral memory.

Documentation should include measurements, CAD drawings, photographs, process videos, test results, assembly notes, parts lists, diagrams, cable maps, media tests, puppet-fabrication notes, audio-installation records, and future recommendations. This material should ultimately be organized for use in GitHub, grant reporting, onboarding, future fabrication, and classroom activity.

5. Student Researcher Roles and Project Participation

Student researchers should be assigned to bounded production-development tasks that contribute to the larger SAR build without requiring them to function as lead engineers. The summer work should support participation across CAD documentation, fabrication support, projection testing, puppet-material studies, media preparation, audio/visual documentation, GitHub integration, and digital-twin support.

Where possible, each student role should connect practical work to a documented research question. For example, a fabrication task may also test whether a particular flat design is repeatable; a projection test may also evaluate front/rear projection behavior; a camera setup may also document mediation geometry; and a puppet-material study may also assess shadow quality, translucency, durability, and camera capture.

The intended outcome is a collaborative production-research environment in which students contribute to the construction of SAR while also producing reusable documentation for future cohorts. Student work should therefore produce visible artifacts: diagrams, test matrices, photographs, process videos, CAD files, assembly notes, media samples, or GitHub documentation.

6. Proposed Summer Roadmap Beginning July 10

The following schedule assumes a start date around July 10 and continuing work into August. It is intentionally high-level and should be refined once staffing, procurement arrival dates, room access, and fabrication constraints are confirmed.

Phase	Dates	Primary Focus	Key Activities	Likely Deliverables
1. Orientation, GitHub Integration, and Build Preparation	July 10–16	Establish shared documentation and prepare the first 8-foot build	Review project goals; inspect LG-038; integrate existing room and prototype measurements into GitHub; document 6-foot prototype as reference; review SeaChange 360 fabric findings; inventory available materials; identify parts needed for first 8-foot flat	GitHub measurement package; photo archive; initial room and triptych diagrams; first 8-foot flat build plan; draft test plan
2. First 8-Foot Flat Construction and Parallel Media Tests	July 17–24	Build a revised 8-foot working prototype while continuing projection/media experiments	Begin construction of one 8-foot flat; use the 6-foot triptych as a test rig where useful; test projection fabrics; compare front/rear projection; evaluate initial tensioning methods; observe masking needs; test camera capture; begin layered-video shadow tests; review past performance recordings for projection use	First 8-foot flat assembled or substantially underway; fabric/projection test matrix; tensioning notes; projection photographs; preliminary camera notes; initial layered-video tests; candidate prerecorded projection excerpts
3. Revised Flat Evaluation and Standard Flat	July 24–31	Learn from the first 8-foot flat and define repeatable architecture	Evaluate handling, stability, scale, puppeteer usability, and projection behavior of the first 8-foot flat; select preferred fabric attachment strategy; define frame dimensions; identify hinge/pivot logic; reserve speaker/sensor/power zones; create CAD drawings	First-flat evaluation notes; standardized flat design draft; parts/cut list; masking concept; power/cable-routing concept

Phase	Dates	Primary Focus	Key Activities	Likely Deliverables
Design				
4. 8-Foot Triptych Fabrication and Puppet-Material Studies	July 31 – August 14	Build larger structures while developing related fabrication tests	Assemble 8020 frames; install hinge/pivot systems; test bases/wheels; install or mock up fabric tensioning; refine masking strategy; begin puppet fabrication tests using purchased materials; develop preliminary puppet-stage module concepts, including depth requirements and organza-roll options	One or more 8-foot triptych structures assembled or partially assembled; fabrication documentation; revised drawings; puppet-material test notes; preliminary puppet-stage module concept
5. Media, Audio, and Digital-Twin Integration Testing	August 14 – 23	Test projection, camera, audio, and room-representation assumptions	Project onto completed surfaces; test camera views; evaluate acoustic-transparency options; consider top/rear speaker placement; document projector geometry; document immersive audio installation/configuration; update digital-twin measurements and equipment locations	Projection/camera geometry notes; speaker-placement recommendations; updated test matrix; media samples; audio-installation notes; digital-twin update package
6. Stabilization and	Augus	Package results for fall continuation	Finalize diagrams; edit process documentation; summarize unresolved issues; create assembly and operation notes; identify fall priorities	Triptych Flat System v1 package; installation documentation; future work list; student-facing onboarding materials

Phase	Dates	Primary Focus	Key Activities	Likely Deliverables
Documentation	23–31			

7. Expected End-of-Summer Deliverables

The summer should aim to produce a reusable infrastructure package rather than only a finished scenic object. The expected deliverables include:

- GitHub-integrated measurement documentation for the room and existing prototype;
- CAD or diagrammatic documentation of the standardized flat;
- at least one early 8-foot flat constructed as a revised working prototype;
- documented 8-foot triptych design and assembly process;
- fabric attachment and tensioning findings;
- front/rear projection test results;
- acoustic-transparency observations where possible;
- camera/projection geometry notes;
- preliminary speaker-mounting recommendations;
- masking and visual-continuity recommendations;
- power and cable-routing concept diagrams;
- process photography and video documentation;
- layered-video shadow-casting test results;
- puppet-material fabrication findings;
- preliminary puppet-stage module concept, including panel attachment, depth, and organza-roll/pigmented-material options;
- prerecorded performance projection tests;
- digital-twin measurement and placement records;
- immersive audio installation/configuration notes;
- recommendations for fall continuation and sensor integration.

8. Open Questions for Refinement

Several decisions remain open and should be resolved as the roadmap develops:

- What are the exact target dimensions for each 8-foot flat?
- Should all flats be identical and reversible?
- What hinge or pivot method best balances flexibility and visual continuity?
- Which fabric attachment method produces the best combination of flatness, replacement, and repeatability?
- Should the projection surface sit near the front plane, rear plane, or be adjustable?
- How much masking is acceptable before the triptych feels visually fragmented?
- Should each flat reserve space for one speaker, and should that speaker be front-mounted, top-mounted, or behind the fabric?
- Which projection fabrics from the SeaChange 360 analysis should be retested?
- What level of power distribution can be safely completed this summer?
- What must be completed by August for the room to be useful in fall development?

9. Working Principle

The guiding principle for the summer should be to build a modular system that can evolve. The immediate task is to fabricate and test triptychs, but the deeper goal is to establish a reusable physical-media language for LG-038. If the flat standard is successful, it can support projection, masking, localized sound, cameras, sensors, future AI-mediated observation, and student-led research projects without requiring the system to be rebuilt from scratch each semester.